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30 SEPTEMBER 1958
SUPERSEDING
MIL-C-10463A
4 JUNE 1952

MILITARY SPECIFICATION

CHEMICAL AGENT, CYANOGEN CHLORIDE

This specification has been approved by the Department of Defense and is mandatory for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of cyanogen chloride (CNCl).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-T-3250 — Tank, Liquid Gas, One-Ton, Type D.

MIL-T-10451 — Tetrasodium Pyrophosphate; Anhydrous, Technical (for Stabilizer).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition—Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

FSC 1365

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3. REQUIREMENTS

3.1 **Material.** The material shall conform with the requirements of table I.

TABLE I. *Requirements*

Requirement	Percent	Test method
Water, max.	0.25	4.3.1
Residue on evaporation, max.	0.02	4.3.2
Acidity (as HCl), max.	0.024	4.3.3
Free chlorine, max.	0.005	4.3.4
Cyanogen chloride, min.	96.0	4.3.5
Hydrogen cyanide, max.	3.0	4.3.6
Iron, max.	0.002	4.3.7

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Inspection provisions.**

4.2.1 *Lotting.* A lot shall consist of that quantity of cyanogen chloride produced by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no changes in materials, provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot.

4.2.1.1 *Batch.* A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

4.2.2 *Sampling.*

4.2.2.1 *For examination.* Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.2.2.2 *For test.* A sample shall be taken from each lot, the size to be calculated on the basis of one tenth of the square root of the number of containers in the lot raised to the next highest whole number. If there are fewer than three containers in a lot, each container shall be sampled. In all other cases, no fewer than three containers shall be selected. From each container in the sample a representative one pound specimen shall be taken and placed in separate, clean, dry containers which are labeled to identify the lot and container represented.

4.2.3 *Examination.*

4.2.3.1 *Nondestructive.* Each lot shall be inspected in accordance with the classification of defects and Standard MIL-STD-105.

4.2.3.2 *Tests.* Equal portions of each specimen selected in accordance with 4.2.2.2 shall be thoroughly mixed to form a composite specimen. The composite specimen shall be subjected to the tests specified in 4.3.

4.2.3.2 *Classification of defects.*

4.2.3.2.1 *Preparation for delivery. Categories and defects.*

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Major—AQL 2.5 percent defective

1. Container incorrect.
2. Container contaminated.
3. Container damaged or leaking.
4. Proportionate quantities of chemical agent packed per container incorrect.
5. Container closure incorrect.
6. Marking incorrect, missing or illegible.

4.3 Tests. Distilled water and analytical reagent grade chemicals shall be used throughout the test. When applicable blank determination shall be run and corrections applied where significant.

4.3.1 *Water.* Dry a vaporizing bulb (D of figure 1) at 100° ± 2°C., cool in a desiccator, and weigh to the nearest milligram (mg.). Close the inlet and outlet of the bulb by means of rubber tubing sealed at one end with a small piece of glass rod. Cool the bulb in an ice bath and introduce about 10 milliliters (ml.) of the sample (free from suspended matter) into the bulb and reweigh to the nearest mg. Pinch off both sections of rubber tubing with Hoffman clamps and assemble in the train as shown in figure 1. Apply gentle suction to avoid sucking the solution out of the absorption columns (B_1 and B_2). Slowly open stopcocks of the tared phosphorus pentoxide drying tube and loosen the Hoffman clamps on the vaporizing bulb. Open drying tubes E to the atmosphere. Regulate the flow so that the liquid cyanogen chloride does not solidify. When the cyanogen chloride has evaporated, immerse the vaporizing bulb in a water bath at 55° ± 5°C. for about 5 minutes to insure complete evaporation of the water. Continue passing air through the train for an additional 5 minutes. Disconnect the apparatus and reweigh the phosphorous pentoxide drying tube and the dried vaporizing bulbs to the nearest mg. Reserve the sodium hydroxide solution in the columns (B_1 and B_2) for the determination of cyanogen chloride and hydrogen cyanide (see 4.3.5 and 4.3.6). Calculate the water content as follows:

$$\text{Percent water} = \frac{100 \times (A - B)}{W}$$

where:

A = weight of drying tube after evaporation of sample, in grams (gm.).
B = weight of drying tube before evaporation of sample, in gm.
W = weight of specimen used (wet cyanogen chloride), in gm.

4.3.2 *Residue on evaporation.* Calculate the percentage of nonvolatile residue as follows:

$$\text{Percent nonvolatile residue} = \frac{100 \times (C - D)}{W}$$

where:

C = weight of vaporizing bulb after volatilization. in gm.
D = weight of vaporizing bulb (tare), in gm.
W = weight of specimen used, in gm.

4.3.3 *Acidity (as HC1).* Place 150 ml. of water in a 350-ml. magnesia pressure bottle, add 2 drops of saturated solution of methyl red, and cool to 0° to 1°C. in an ice bath. Add 5 ml. of the sample. Close the bottle, shake vigorously for 15 seconds, and titrate at once with 0.02 *N* sodium hydroxide solution. Add the sodium hydroxide at a rapid rate until a yellow color is obtained which persists for approximately 5 seconds. Calculate the percent acidity as HC1 as follows:

$$\text{Percent acidity as HC1} = \frac{3.6 \times V \times N}{W}$$

where:

V = volume of sodium hydroxide solution used, in ml.
N = normality of sodium hydroxide solution.

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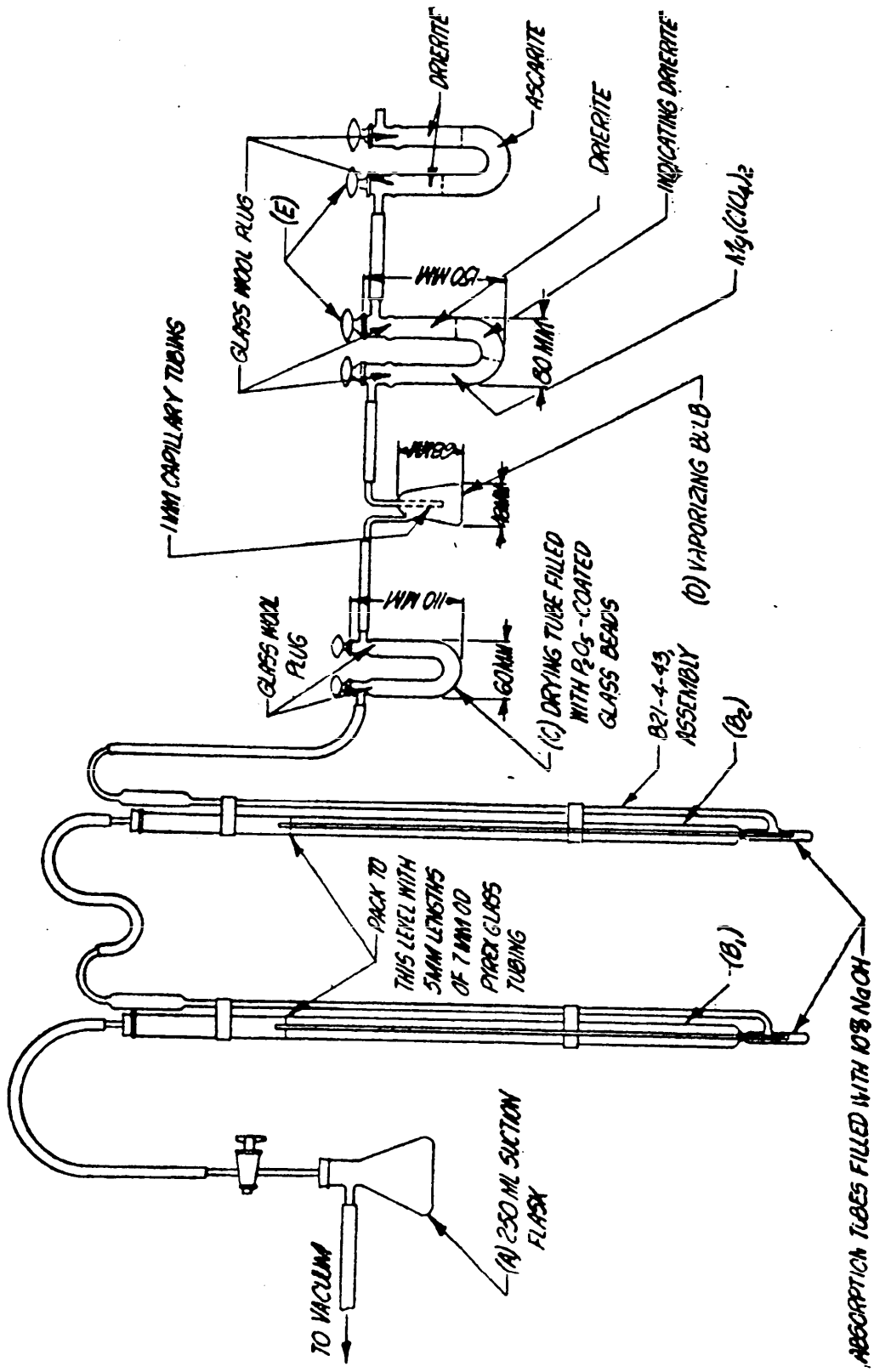


FIGURE 1. *Apparatus for CNCL determination.*

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W = weight of specimen used, in gm.

4.3.4 *Free chlorine.* Dissolve 2 gm. of potassium iodide in 150 ml. of water in a 350-ml. magnesia pressure bottle, and cool to about 0°C. in an ice bath. Add 5 ml. (specific gravity 1.2) of the specimen (a 10-ml. graduate is suitable), shake thoroughly, add 100 ml. of a cold 10-percent solution of sodium hydroxide, and shake again. Cool in the ice bath to about 0°C., transfer to a 750-ml. Erlenmeyer flask, acidify with a cold 1 to 2 solution of sulfuric acid, and titrate at once with 0.05 *N* sodium thiosulfate using starch solution as the indicator. Caution should be observed to prevent mistaking a brown coloration for the starch-iodine reaction. Calculate the percent free chlorine as follows:

$$\text{Percent free chlorine} = \frac{3.5 \times V \times N}{W}$$

where:

V = volume of sodium thiosulfate solution used, in ml.

N = normality of sodium thiosulfate solution.

W = weight of specimen used, in gm.

4.3.5 *Cyanogen chloride content.* Transfer quantitatively the sodium hydroxide solutions from columns B_1 and B_2 (see 4.3.1) to a 2-liter volumetric flask. Dilute to the mark and mix thoroughly. Transfer a 50-ml. aliquot to a 600-ml. beaker. Reserve the residual portion of the solution for the hydrogen cyanide determination. Add 5 ml. of 37-percent formaldehyde solution (the formaldehyde is added to prevent interference by the cyanide). Stir the solution, let stand 15 minutes, dilute to about 200 ml. neutralize with nitric acid and add 5 ml. in excess. Determine the total chlorine by the Volhard method. Calculate the percent cyanogen chloride as follows:

$$\text{Percent cyanogen chloride} = \frac{6.148 \times (VN - BM)}{W} - 1.7 \times (C + D)$$

where:

V = volume of silver nitrate solution used, in ml.

N = normality of silver nitrate solution.

B = volume of potassium thiocynate used in back titration, in ml.

M = normality of potassium thiocyanate solution.

C = Percent hydrochloric acid (see 4.3.3).

D = percent free chlorine (see 4.3.4).

W = weight of specimen in aliquot used, in gm.

4.3.6 *Hydrogen cyanide.* Transfer a 500-ml. aliquot of the solution reserved for this determination (see 4.3.1) to an 800-ml. beaker and neutralize with nitric acid using litmus paper as the indicator. Immediately add 1 ml. of ammonium hydroxide and 0.2 gm. of potassium iodide for each 100 ml. of the solution. Stir to dissolve the potassium iodide. Place the beaker on a black background and titrate with 0.1 *N* silver nitrate solution, with continual stirring, to a faint permanent turbidity. Calculate the percent hydrogen cyanide as follows:

$$\text{Percent hydrogen cyanide} = \frac{5.4 \times V \times N}{W}$$

where:

V = volume of silver nitrate solution used, in ml.

N = normality of silver nitrate solution.

W = weight of specimen in aliquot used, in gm.

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4.3.7 *Iron.*

4.3.7.1 *Colorimetric standards.* Dissolve 7.02 gm. of ferrous ammonium sulfate ($FeSO_4(NH_4)_2SO_4.6H_2O$) in about 100 ml. of water in a 500-ml. beaker. Add 5 ml. of sulfuric acid (specific gravity 1.84) and 1 ml. of bromine. Boil gently until the excess bromine is expelled. Cool and transfer the solution quantitatively to a 1-liter volumetric flask. Dilute to volume with water and store and use as needed. This stock solution keeps indefinitely and contains 1 mg. of iron per ml. Dilution of the stock solution, 1 to 100, gives a solution of suitable strength for the preparation of the color standards. This dilute stock solution should be prepared weekly. Transfer suitable volumes of the dilute stock solution (containing 0.01 mg. of iron per ml.) to 50-ml. Nessler tubes containing about 10 ml. of water. Add 2 ml. of nitric acid (specific gravity 1.42), 2 ml. of a 20-percent solution of ammonium thiocyanate, and 30 ml. of acetone. Dilute to the mark with water and mix thoroughly. Keep the standards, when not in use, in a dark place. Prepare fresh standards every 8 hours.

4.3.7.2 *Procedure.* Pour 5 ml. of the sample into a 50-ml. beaker and allow to evaporate under a hood (slight warming permitted). Add 2 ml. of nitric acid (specific gravity 1.42), cover the beaker with a watch glass and heat (do not boil) for 2 to 3 minutes. Transfer the contents of the beaker, quantitatively, to a 50-ml. Nessler tube. Add 2 ml. of a 20-percent solution of ammonium thiocyanate and 30 ml. of acetone. Dilute to the mark with water, mix thoroughly, using a clean stopper, and compare with the standards prepared as described above.

4.4 **Rejection criteria.** If composite specimen selected as specified in 4.2.3.2 fails to conform to this specification, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**

5.1.1 *Level A.* The chemical agent shall be packed in clean one-ton steel containers conforming to Specification MIL–T–3250. Before introducing the cyanogen chloride into the container, a stabilizer consisting of seventy-nine and one-half plus or minus one and one-half (79.5 ± 1.5) pounds of anydrous tetrasodium pyrophosphate conforming to the requirements of Specification MIL–T–10451 shall be introduced into the container. After the stabilizer has been put into the containers, one thousand six hundred plus or minus thirty (1600 ± 30) pounds of cyanogen chloride shall then be packed into the steel containers.

5.1.2 *Levels B and C.* Not applicable.

5.2 **Marking.** In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129 and I. C. C. Regulations. Markings shall also include the manufacturer's lot number and date of manufacturer of the chemical agent. Containers containing stabilized cyanogen chloride shall be further marked on the valve end with "+P" (at least 1 inch high.)

5.2.1 *Precautionary marking.* All containers shall be marked with the following:

DANGER: POISONOUS LIQUID

Do not breathe vapors.
Avoid contact with skin.
Do not drop container.
Store in a cool place.

POISON

In case of accident get first aid or call physician immediately.

6. NOTES

6.1 **Intended use.** The cyanogen chloride covered by this specification is intended for use as chemical agent.

6.2 **Ordering data.** Procurement documents should specify the title, number, and date of this specification.

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Custodians:

Army—Chemical Corps
Navy—Bureau of Ordnance
Air Force

Preparing activity:

Army—Chemical Corps

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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